Reconstruction Without Pairwise Distances:
A Large-Scale Comparison of Autoencoders, DiRe, and UMAP

Igor Rivin

17 July 2026

Abstract

Graph-based dimension-reduction methods explicitly construct and optimize relationships between observations. A reconstruction autoencoder instead learns from individual input vectors using only mean-squared error. We compare these approaches at target widths $d \in \{2, 8\}$ on 723,457 arXiv paper embeddings and one million Wikipedia paragraph embeddings. All reducers receive the same row-normalized data and are evaluated over three seeds using sample-conditional neighborhood fidelity, sampled global-distance fidelity, local decodability, label- or group-aware diagnostics, sampled persistence-derived diagnostics, and synchronized fit time. We add a separate single-cell arm on 100,000 deterministically selected Tabula Sapiens cells: PCA, the autoencoder, DiRe, and UMAP reduce the same unnormalized 50-PC prepared matrix, and an exact-neighbor audit tests two DiRe NN-descent working-degree settings. We separately test one-pass $96 \to 8$ scaling on exact 10-million- and 100-million-row prefixes of Yandex Deep1B, with bounded-memory training, exact reduced-space retrieval, and sampled topology diagnostics. Controlled 10-million-row experiments then separate distinct training examples from optimizer exposure and trace training through 9,632 updates.

No method dominates the completed cells. At arXiv width two, the autoencoder leads the rank-aware local scores, UMAP gives the best category prediction, and DiRe gives the lowest sampled Betti-curve discrepancies. At width eight, the autoencoder leads all six primary neighborhood, distance, and decoding scores while UMAP still gives the best category prediction. On the Wikipedia corpus, width two divides the geometric columns among all three methods, while at width eight the autoencoder leads the local, distance-order, and decoding scores but not stress; DiRe gives the lowest global-distance stress at both widths and UMAP most strongly groups paragraphs from the same article. On Tabula Sapiens, UMAP gives the strongest donor-held-out cell-type scores at both widths, while the autoencoder leads four of five PCA50-relative geometric scores at width eight. Increasing DiRe's NN-descent working degrees from 31/62 to 64/128 raises exact-sample Recall@30 only from .99585 to .99601; the absolute change in every plotted downstream mean is below .0020. A separate controlled check on manifolds of known dimension finds that rounded TwoNN and autoencoder-elbow estimates agree exactly in four of nine seeded trials and are within one dimension in all nine, with interpretable differences between local and global reconstructive dimension.

On Deep1B, the autoencoder gives bounded-state, approximately linear one-pass processing from 10 to 100 million rows. UMAP completes at 100 million with 78.34 GB of sampled total GPU-process memory, while the tested DiRe fit fails on a 102.4 GB allocation. Within the fixed 10-million-row control, increasing distinct training rows from 65,536 to 9.86 million at a common 1,204-update budget changes legacy-evaluation NMSE by only 0.20%; observed pool differences are therefore small under this short protocol. Yet a frozen tune-only sweep finds no qualifying plateau under its prespecified NMSE flatness rule through 9,632 updates and shows a descriptive pool-by-budget separation: the one-million-row pool ends at lower tune NMSE than the 65,536-row pool (.3956 versus .4060). Strict-ID recall remains low—Recall@10 and Recall@100 are .0358 and .0677 for the 10-million-row one-pass layout—so systems scalability is not nearest-neighbor adequacy. These are implementation- and protocol-specific observations.

The sampled topology comparison is similarly conditional on corpus and width. DiRe leads both Betti-curve discrepancies for arXiv at width two; at width eight the autoencoder leads β_0 and DiRe leads β_1. On Wikipedia, DiRe leads β_0 and the autoencoder leads β_1 at width two, while the autoencoder leads both at width eight. A paired BF16-autocast sensitivity check preserves reconstruction and shared-metric quality under prespecified practical margins. It nearly halves Wikipedia training time, but slightly slows the smaller arXiv network, showing that mixed-precision speedup depends on whether dense computation amortizes its overhead.

Keywords: dimensionality reduction; autoencoders; manifold learning; intrinsic dimension estimation; neighborhood preservation; large-scale machine learning; GPU acceleration.

1 Introduction

Dimension reduction is not a single optimization problem. Given observations $X = \{x_i\}_{i=1}^N \subset \mathbb{R}^D$ and a target width $d \ll D$, a reducer must discard information, but different objectives specify different information as expendable. Coordinate reconstruction rewards recovery of each input vector. A neighborhood objective rewards the survival of selected short-range relationships. A force layout can instead balance graph-edge attraction against broad repulsion. None of these criteria alone implies preservation of global distances, labels, density, or persistent homology.

This paper compares one representative of each objective at fixed target widths: a reconstruction autoencoder, UMAP, and DiRe. The comparison is deliberately broader than a two-dimensional visualization contest. We evaluate $d = 2$, where severe compression exposes clustering and topological behavior, and $d = 8$, where a reducer has more room to preserve geometry for downstream computation. Text embeddings, single-cell data, and the Deep1B scale pilot then test whether conclusions survive changes in ambient dimension, sparsity history, labels, and corpus size.

1.1 Three algorithms, three fitted objects

Method	Object optimized	Input relationships used during fitting	Result of a completed fit
AE	Encoder and decoder weights under reconstruction MSE	Individual vectors without a graph or pairwise loss	A parametric map $f_\theta(x)$ and decoder $g_\phi(z)$
UMAP	Low-dimensional coordinates under fuzzy-graph cross-entropy	Locally rescaled, weighted k-NN graph plus sampled nonedges	A transductive layout whose new points require UMAP's separate transform procedure
DiRe	Low-dimensional coordinates under bounded attractive and repulsive forces	k-NN adjacency plus freshly sampled repulsive partners	A transductive force-directed layout in the implementation tested here

Table 1: Algorithmic contrast. The table describes the objectives used in this study, not every extension available for each method.

The table separates two questions that are often conflated. The first is what statistical object is fitted. The second is how the implementation obtains enough relationships to fit it. UMAP and DiRe store only a sparse $O(Nk)$ graph after neighbor construction, but that graph may be built by exact tiled search or by an approximate index. The autoencoder has no graph-building stage. Its batched training state is bounded by the model and optimizer, although materializing or saving all N encoded rows still requires $O(Nd)$ output storage.

1.2 The reconstruction autoencoder

An autoencoder is the simplest algorithm in this comparison [1]. An encoder $f_\theta : \mathbb{R}^D \to \mathbb{R}^d$ maps an observation to its bottleneck coordinates, and a decoder $g_\phi : \mathbb{R}^d \to \mathbb{R}^D$ maps those coordinates back to input space. Our model minimizes only elementwise reconstruction error,

$$\mathcal{L}_{\mathrm{AE}}(\theta, \phi) = \frac{1}{ND} \sum_{i=1}^{N} \|x_i - g_\phi(f_\theta(x_i))\|_2^2 . \tag{1}$$

There is no label, pairwise-distance, neighborhood, contrastive, or topology term. The nonlinear hidden layers allow a curved encoding, while the linear bottleneck and output leave the latent coordinates and reconstructed values unconstrained in sign.

For a fixed architecture, batch size, and epoch count, the dominant dense work is

$$O\left(EN \sum_{\ell} w_\ell w_{\ell+1}\right),$$

where E is the number of epochs and w_ℓ are adjacent layer widths. The dependence on N is linear, and the matrix multiplications map efficiently to GPU tensor cores. This makes bounded-memory streaming natural: batches may be discarded after an optimizer update, and the trained encoder subsequently processes new rows without rebuilding a graph. The benefit is conditional. MSE is coordinate- and scale-sensitive, and a model that reconstructs high-variance directions well need not preserve nearest neighbors, class boundaries, or topology. We also train a fresh model for every corpus, target width, and seed, so the experiments do not assume a universal reusable encoder.

1.3 UMAP: a fuzzy weighted-neighbor objective

UMAP constructs a weighted graph by assigning a locally adapted scale to every observation [2]. Let $\mathcal{N}_i$ contain the k nearest neighbors of x_i, let d_{ij} denote their input-space distance, and let ρ_i be the distance to the nearest neighbor. UMAP chooses $\sigma_i > 0$ so that the fuzzy cardinality is approximately $\log_2 k$, then defines the directed membership

$$p_{j|i} = \exp\left[-\frac{\max(0, d_{ij} - \rho_i)}{\sigma_i}\right], \qquad j \in \mathcal{N}_i. \tag{2}$$

Membership is zero outside the computed neighborhood. The probabilistic fuzzy union is

$$p_{ij} = p_{j|i} + p_{i|j} - p_{j|i}p_{i|j}. \tag{3}$$

This turns the directed neighborhoods into a symmetric weighted graph. Local rescaling is the defining distinction from an ordinary unweighted k-NN graph: equal ambient distances can receive different memberships in dense and sparse regions.

For candidate coordinates y_i, UMAP models low-dimensional membership by

$$q_{ij} = \left(1 + a\|y_i - y_j\|_2^{2b}\right)^{-1}. \tag{4}$$

Parameters a and b are fitted from `min_dist` and `spread`. The formal layout objective is fuzzy-set cross-entropy,

$$C_{\mathrm{UMAP}} = \sum_{i \neq j} \left[p_{ij} \log \frac{p_{ij}}{q_{ij}} + (1 - p_{ij}) \log \frac{1 - p_{ij}}{1 - q_{ij}}\right]. \tag{5}$$

The first optimized term attracts graph edges, whereas the second repels pairs that should not be edges. Enumerating every non-edge would restore quadratic work, so the practical algorithm samples positive edges according to their membership and uses negative sampling for repulsion. A spectral embedding of the fuzzy graph supplies the usual initial coordinates, after which a linearly cooled stochastic optimizer refines them.

Our implementation is the GPU UMAP in RAPIDS cuML 26.06 [6]. It uses Euclidean input distance, `min_dist`= 0.1, `spread`= 1, the default five negative samples per positive, and 200 epochs for these large inputs. The text and single-cell comparisons use spectral initialization and force the serial-epoch kernel, which cuML documents as the stable seeded path for that initialization. Deep1B instead uses random initialization and cuML's automatic epoch-kernel choice. The neighbor builder is also experiment-specific. The arXiv and Wikipedia runs request `build_algo=auto` with a fixed `random_state`, and cuML 26.06 resolves that combination to exact brute-force k-NN. The Tabula Sapiens and Deep1B runs explicitly request cuVS NN-descent [5, 7]. In those arms, NN-descent avoids an all-pairs search by iteratively improving a candidate neighbor graph, at the cost of an additional accuracy and nondeterminism control.

1.4 DiRe: direct forces on a neighbor graph

DiRe is a three-stage reducer [3, 4]. It constructs a k-NN graph, initializes low-dimensional coordinates by PCA, a spectral method, or a random projection, and then applies a force-directed layout. The experiments here use PCA initialization. Unlike UMAP's fuzzy union, the tested PyTorch/RAPIDS layout uses the neighbor indices as an unweighted attraction graph: input-space distances select the neighbors, but do not weight their subsequent attractive forces.

The operational force law is useful for understanding both the method and its speed. At layout iteration t, set $r_{ij} = \|y_j - y_i\|_2$ and $u_{ij} = (y_j - y_i)/(r_{ij} + \varepsilon)$. For neighbor set $\mathcal{N}_i$ and a fresh uniform negative sample $\mathcal{S}_i^{(t)}$, the implementation computes

$$A(r) = \frac{r^{2b}}{r^{2b} + a}, \qquad R(r) = \frac{\exp(-r/c)}{1 + ar^{2b}}, \tag{6}$$

$$F_i^{(t)} = \sum_{j \in \mathcal{N}_i} A(r_{ij}) u_{ij} - \sum_{s \in \mathcal{S}_i^{(t)}} R(r_{is}) u_{is}, \qquad y_i^{(t+1)} = y_i^{(t)} + \left(1 - \frac{t}{T}\right) \mathrm{clip}_c(F_i^{(t)}). \tag{7}$$

As in UMAP, a and b are fitted from `min_dist` and `spread`. The exponential factor and componentwise clipping bound distant repulsion and individual updates. The default negative-sample ratio is eight per neighbor, and the completed layouts use $T = 128$ iterations. Thus, once the graph exists, a fixed-k layout iteration has $O(Nk(1+\gamma)d)$ arithmetic for negative ratio γ, with no all-pairs layout force.

The `dire-rapids` implementation builds neighbors with cuVS and evaluates the layout in vectorized PyTorch. Its fused force kernel uses BF16 for the gather-heavy force calculation, accumulates positions in FP32, and falls back to chunked evaluation if the full vectorized step does not fit. The text runs use the 0.3.1 index-search path, which selected IVF-PQ for arXiv and IVF-Flat for Wikipedia. The single-cell arm uses the API-compatible 0.3.2 development path and cuVS all-neighbors NN-descent. The working-degree audit below tests whether that newer graph builder changes downstream quality.

UMAP and DiRe therefore share a family resemblance, not an objective. Both attract neighbors and approximate broad repulsion with negative samples. UMAP first converts distances into an edge-weighted fuzzy union and samples those edges while optimizing cross-entropy. DiRe applies a direct, synchronous bounded force to every stored neighbor at every iteration. Neither implementation optimizes a persistence loss. DiRe's topology-preservation motivation is evaluated here as an empirical question, using the same sampled persistence diagnostics applied to all methods.

1.5 Questions addressed by the experiments

The study asks four linked questions:

(i) Which objective best preserves local neighborhoods, sampled global distances, local decodability, labels or groups, and sampled topology at $d = 2$ and $d = 8$?
(ii) How do synchronized fit time and measured memory change from medium corpora to 10-million- and 100-million-row Deep1B prefixes?
(iii) Does the autoencoder reconstruction elbow give an intrinsic-dimension estimate comparable to local TwoNN and maximum-likelihood estimators?
(iv) Are the conclusions sensitive to BF16 training, autoencoder exposure and training-pool size, or DiRe NN-descent working degree?

The next section fixes the data and implementations. We then define every metric before reporting results, so a method is not declared successful merely because it optimizes a related quantity. The conclusion separates completed empirical comparisons from systems observations and from hypotheses that would require further experiments.

2 Data, reducers, and protocol

Corpus	Rows	Input D	Target d	Label field	Label interpretation
arXiv	723,457	384	2, 8	primary category	subject category
Wikipedia	1,000,000	1,024	2, 8	article URL	same-article group
Tabula Sapiens	100,000	50	2, 8	cell type	donor-held-out annotation

Table 2: The three fixed-width benchmark inputs. The first two are row-normalized text embeddings; Tabula Sapiens is an unnormalized, non-whitened 50-PC single-cell representation. Wikipedia URLs are group identifiers, not a topic taxonomy analogous to arXiv categories or Tabula cell-type annotations.

The arXiv matrix contains mean-pooled `BAAI/bge-small-en-v1.5` paper embeddings [13]. The Wikipedia matrix is the first million rows of the English configuration of `CohereLabs/wikipedia-2023-11-embed-multilingual-v3` [14].

The single-cell arm uses the "Tabula Sapiens—All Cells" dataset from CELLxGENE Census release 2025-11-08 [17, 18]. Exactly 100,000 cells are selected by deterministic donor-by-cell-type stratified sampling from 1,136,218 eligible primary cells. Genes detected in fewer than three selected cells are removed. On raw counts, 2,000 Seurat-v3 highly variable genes are selected with donor-by-assay batches; counts are normalized to 10,000 per cell, transformed by `log1p`, centered, scaled to unit variance, clipped at ten, and reduced to 50 non-whitened principal components. No additional cell filter or row normalization is applied.

Every text-embedding row is converted to float32 and normalized to unit Euclidean norm. Every single-cell reducer instead receives the same prepared float32 PCA50 values verbatim and in the same row order. All methods receive the complete corresponding matrix at both $d = 2$ and $d = 8$, with reducer/training seeds 0, 1, and 2. Evaluation samples and sampled pairs are held fixed across methods, target widths, and seeds. DiRe and UMAP fitting, and neighborhood-based evaluation, use $k = 16$ for the text corpora and $k = 30$ for Tabula Sapiens; autoencoder fitting has no neighborhood parameter. The comparisons are transductive: every reported layout is fitted on the full corpus being evaluated.

Autoencoder. Training minimizes only the elementwise reconstruction MSE in Eq. (1). Hidden layers use SiLU activations; bottleneck and output layers are linear. The architectures are

$$384 \to 256 \to 128 \to 32 \to d \to 32 \to 128 \to 256 \to 384$$

for arXiv and

$$1024 \to 512 \to 256 \to 64 \to d \to 64 \to 256 \to 512 \to 1024$$

for Wikipedia, and

$$50 \to 256 \to 128 \to 32 \to d \to 32 \to 128 \to 256 \to 50$$

for Tabula Sapiens. As d changes from two to eight, the text models' parameter counts range from 271,682 to 272,072 and from 1,346,434 to 1,347,208, respectively. AdamW uses learning rate 10^{-3} and weight decay 10^{-5}, with batches of 32,768 text rows and 4,096 single-cell rows. The text-corpus primary comparison uses float32 arithmetic; the Tabula autoencoder uses BF16 autocast for training and float32 for validation, encoding, and reconstruction evaluation. A separate paired BF16 sensitivity arm for the text models is reported below.

Five percent of the rows select an epoch count under a 200-epoch cap and patience 15. A freshly initialized model is then refit on all rows for the selected number of epochs. The autoencoder is neither pretrained nor recycled between corpora, target widths, or seeds.

DiRe and UMAP. The original text-corpus runs use DiRe 0.3.1's cuVS implementation, with PCA initialization, 128 layout iterations, and an automatically selected approximate-neighbor index. The Tabula arm uses the API-compatible local DiRe 0.3.2 development implementation; under its automatic RAPIDS 26.06 path, cuVS all-neighbors NN-descent constructs the graph with PCA initialization and the same 128 layout iterations. Exact development identifiers remain in the machine-readable experiment record rather than the public narrative. cuML UMAP 26.06 uses Euclidean distance, spectral initialization, `min_dist`= 0.1, and `spread`= 1. The seeded text runs request `build_algo=auto`, which resolves to brute-force k-NN in this cuML release; the single-cell and Deep1B arms explicitly request NN-descent. Native UMAP inverse transformation is disabled because its CPU Delaunay stage is not a scale-safe operation for these corpora.

Hardware and timing. Reducer runs use a single NVIDIA GH200, RAPIDS/cuML 26.06, PyTorch 2.11, and three independent reducer/training seeds. Persistence calculations use Ripser 0.6.14 and FastDTW 0.3.4 in the RAPIDS 26.04 environment because those packages were unavailable in the 26.06 environment. CUDA synchronization brackets every reported fit. DiRe and UMAP time their synchronized `fit_transform`; autoencoder time includes GPU staging, validation-based training, fresh full-data refitting, full-corpus encoding, and native reconstruction statistics. Source loading, row normalization, library import, downstream comparison metrics, and the final layout-file flush are outside "fit." One-time data preparation took 2.58 seconds for arXiv and 10.05–14.59 seconds across the two Wikipedia runs; it may be reused across target widths.

Single-cell graph audit. At fixed final $k = 30$, DiRe's default all-neighbors configuration uses NN-descent working graph degree 31 and intermediate degree 62. We compare it with the UMAP-like degrees 64 and 128 while holding 20 NN-descent iterations, termination threshold 10^{-4}, input, layout settings, and seeds fixed. Each configuration has three independent graph builds. Exact cuVS brute-force search queries the same 5,000 deterministic cells against all 100,000 input rows, and each saved approximate graph is then reused for paired $d = 2$ and $d = 8$ DiRe layouts. The second configuration matches working degrees inside DiRe's cuVS path; it does not extract or reuse UMAP's graph.

Deep1B scaling pilot. The separate scaling experiment uses exact leading 10-million- and 100-million-row prefixes of Yandex Deep1B, whose one billion 96-dimensional GoogLeNet image descriptors were PCA-compressed, row-normalized, and released with public queries and Euclidean nearest-neighbor ground truth [15, 16]. Every pilot layout has $d = 8$. The autoencoder is

$$96 \to 256 \to 128 \to 32 \to 8 \to 32 \to 128 \to 256 \to 96,$$

with 124,328 parameters, batch size 8,192, and the same MSE, AdamW learning rate, and weight decay as above. One pass is made over each local-file prefix; a fixed uniform 65,536-row validation sample is excluded from training. Bounded `preadv` blocks feed the model without persistently mapping the corpus. The layout is produced in a second bounded pass, so "streaming" here means bounded local-file access, not a network stream or a single read of each row.

Deep1B training-pool and update-budget controls. The training-pool control fixes the 10-million-row source universe and the architecture, optimizer, batch size, and deterministic FP32 arithmetic above. The original 65,536-row holdout becomes a fixed legacy evaluation cohort; a new disjoint 65,536-row tune cohort selects checkpoints. Five training pools—65,536, 262,144, 1,048,576, 4,194,304, and 9,863,168 rows—are nested prefixes of one uniformly shuffled complement. Seeds 1729–1731 pair initial weights across pools. Every fixed-budget arm receives exactly 1,204 updates, or 9,863,168 row presentations, while a separate checkpoint records the end of the first pool pass. Tune NMSE is inspected on a common 32-update grid and the legacy cohort is evaluated only after tune-only selection. Geometry uses fixed legacy subsets, and strict retrieval is a descriptive seed-1729 spot check for three pools.

The update-budget discovery follows uninterrupted trajectories for the 65,536- and 1,048,576-row pools under the same three seeds. It uses only the tune cohort and fixed tune-cohort geometry subsets; the legacy cohort is not materialized. A frozen grid extends through 4,816 updates and, after its terminal region fails a prespecified flatness check, once more through 7,224 and 9,632. Signed within-seed degradation is measured against the terminal checkpoint. Practical margins are 1% relative for reconstruction and local-decoder NMSE, .005 absolute for reconstruction cosine, exact-neighbor overlap, and stress, .002 for trustworthiness and continuity, and .010 for pair-distance Spearman correlation. A candidate must pass every margin, no seed may exceed twice a margin, and the next two checkpoints must also pass. These are engineering tolerances, not formal equivalence bounds. Update 1,204 exactly replays the earlier model states and tune metrics. No discovery checkpoint advances, so the planned fresh-seed legacy confirmation is not run.

At each graph-method scale, UMAP and DiRe receive one seed-0 random-initialized fit with $k = 16$. UMAP uses RAPIDS 26.06 NN-descent with 16 clusters; a 160-cluster control at 100 million rows restores the estimated per-partition size used at 10 million. DiRe 0.3.0 uses its cuVS IVF-PQ path, 128 iterations, and negative-sample ratio eight. The autoencoder uses seed 1729. These are single-run systems observations, not three-seed comparisons. Each completed arm contributes one fitted layout; the failed 100-million-row DiRe arm contributes none. All run on the same GH200, but memory scopes differ: autoencoder GPU memory is PyTorch peak allocated memory, whereas graph-method GPU memory is sampled through NVML as both a complete process total and an increment above a measured baseline; the scaling table uses the latter. Application RSS is a separate process high-water mark. The table captions and discussion preserve these distinctions rather than adding the scopes.

Autoencoder training time includes bounded source reads, local shuffling and batching, host-to-device copies, forward and backward passes, and optimizer updates. Encoding includes a complete source read, hashing, forward passes, and layout writing; the broader AE total also includes preflight, holdout evaluation, source validation, and public-query encoding. Graph fit is the backend `fit_transform`; graph worker time additionally includes import, bounded input inspection, layout validation, and writing. Download is excluded. Graph RSS can include pages touched through a read-only input memory map, whereas the OS page cache induced by the autoencoder's

positional reads lies outside its process RSS. NVML process memory is sampled every 0.1 seconds. Sequential local-file runs may benefit from uncontrolled page-cache state.

3 Evaluation metrics

Let $X = \{x_i\} \subset \mathbb{R}^D$ be the prepared input—row-normalized for the text corpora and unmodified PCA50 for Tabula Sapiens—and $Y = \{y_i\} \subset \mathbb{R}^d$ a layout. Neighborhood metrics use Euclidean distance and $k = 16$ for the text corpora or $k = 30$ for Tabula. Their $O(n_s^2)$ rank calculations are performed within a declared fixed sample, not the complete corpus: a label-stratified 5,000-paper sample for arXiv, a uniform 3,000-paragraph sample for Wikipedia, and a uniform deterministic 5,000-cell sample for Tabula. Wikipedia has 153,264 URL groups, more than the number of sample slots, so label stratification is not possible. Identical sample indices are used for all methods, target widths, and reducer seeds.

3.1 Neighborhood fidelity

For a sampled point i, let $N_X^k(i)$ and $N_Y^k(i)$ be its k nearest neighbors in input and layout space, and let $r_X(i,j)$ and $r_Y(i,j)$ be the corresponding full within-sample ranks. Define the introduced and lost neighbor sets

$$U_k(i) = N_Y^k(i) \setminus N_X^k(i), \qquad V_k(i) = N_X^k(i) \setminus N_Y^k(i).$$

Trustworthiness [10] penalizes false neighbors introduced by the layout:

$$T(k) = 1 - \frac{2}{n_s k(2n_s - 3k - 1)} \sum_{i=1}^{n_s} \sum_{j \in U_k(i)} \big(r_X(i,j) - k\big). \tag{8}$$

Continuity is reversed trustworthiness and penalizes input neighbors lost by the layout:

$$C(k) = 1 - \frac{2}{n_s k(2n_s - 3k - 1)} \sum_{i=1}^{n_s} \sum_{j \in V_k(i)} \big(r_Y(i,j) - k\big). \tag{9}$$

Both approach one for strong local fidelity. Exact neighbor overlap is

$$O(k) = \frac{1}{n_s k} \sum_{i=1}^{n_s} \left| N_X^k(i) \cap N_Y^k(i) \right|. \tag{10}$$

Unlike trustworthiness and continuity, overlap gives no partial credit when a missed or introduced neighbor has nearly the correct rank.

3.2 Approximate graph recall

For the single-cell audit, let E_i^k be the exact first k input-space neighbors of query cell i and A_i^k the corresponding approximate graph neighbors. On the fixed query set Q of 5,000 cells, we report

$$R_{\text{graph}}(k) = \frac{1}{|Q|k} \sum_{i \in Q} |E_i^k \cap A_i^k|, \qquad F_{\text{exact}}(k) = \frac{1}{|Q|} \sum_{i \in Q} \mathbf{1}[E_i^k = A_i^k \text{ as sets}],$$

for $k \in \{1, 10, 30\}$. The exact search covers the complete 100,000-row prepared input; the approximation is evaluated only on the declared 5% query sample.

3.3 Sampled global distances

For 200,000 fixed, independently sampled non-self pairs P, with pair repetitions allowed, write

$$\delta_{ij} = \|x_i - x_j\|_2, \qquad d_{ij} = \|y_i - y_j\|_2.$$

Pair Spearman correlation ρ_S is the ordinary correlation of the midranks of $\{\delta_{ij}\}$ and $\{d_{ij}\}$. The scale-aligned normalized stress is

$$S = \left[\frac{\sum_{(i,j)\in P}(\delta_{ij} - a d_{ij})^2}{\sum_{(i,j)\in P}\delta_{ij}^2}\right]^{1/2}, \qquad a = \frac{\sum_{(i,j)\in P}\delta_{ij} d_{ij}}{\sum_{(i,j)\in P} d_{ij}^2}. \tag{11}$$

Higher ρ_S and lower S are preferable. Equation (11) is a through-origin least-squares scale alignment; it is not Kruskal Stress-1 or a full Procrustes fit.

3.4 Shared local-decoding probe

A separate sample—10,000 rows for arXiv and 5,000 for Wikipedia—is split 80/20. The training layout is whitened, retaining covariance eigenvectors with $\lambda_j > \max(10^{-8}\lambda_{\max}, \epsilon)$. A distance-weighted 16-neighbor regressor predicts all coordinates of x from the whitened layout. With training and test folds A and B,

$$\text{DecodeNMSE} = \frac{\sum_{i\in B}\|x_i - \widehat{x}(y_i)\|_2^2}{\sum_{i\in B}\|x_i - \bar{x}_A\|_2^2}. \tag{12}$$

This shared transductive probe compares all reducers. It is neither the autoencoder's native decoder error nor an out-of-corpus generalization test.

3.5 Labels, article groups, and cell types

For arXiv, categories occurring at least five times in the fixed metric sample are kept, leaving 4,857 rows and 71 categories. A fixed stratified 80/20 split trains a distance-weighted 16-neighbor classifier in layout space. We report ordinary accuracy and balanced accuracy, the mean per-category recall. Labels are never supplied to a reducer.

For Wikipedia, a URL identifies paragraphs from the same article. The 3,000 fixed anchors query the complete one-million-row layout. Let G_i be the rows sharing anchor i's URL, and let $m_i = |N_Y^{16}(i) \cap G_i|$. Empty and full-corpus-singleton groups are excluded, leaving an eligible set E of 2,895 anchors. We report

$$\text{Purity@16} = \frac{1}{|E|}\sum_{i\in E}\frac{m_i}{16}, \tag{13}$$

$$\text{Hit@16} = \frac{1}{|E|}\sum_{i\in E}\mathbf{1}[m_i > 0], \tag{14}$$

$$\text{TrRecall@16} = \frac{1}{|E|}\sum_{i\in E}\frac{m_i}{\min(16, |G_i| - 1)}. \tag{15}$$

The last quantity is explicitly a *truncated* recall: it can reach one for a large article after retrieving 16 mates, even if additional same-article paragraphs remain.

For Tabula Sapiens, a fixed donor-by-cell-type-stratified 20,000-cell sample supplies a distance-weighted 30-neighbor classifier. Leave-one-donor-out folds prevent a donor's cells from appearing in both the probe's training and test partitions. Cell types must occur at least 20 times across at least three donors; the completed sample evaluates 118 types across 24 donor folds. We report macro-F1 and balanced accuracy. A separate 5,000-cell donor-by-type-stratified sample reports ordinary Euclidean average silhouette width (ASW) over 143 cross-donor cell types; higher values indicate tighter within-type separation. These are post hoc probes of a transductively fitted layout, not tests of reducer generalization to unseen donors.

Technical-covariate leakage uses a separate 20,000-cell sample and, within each eligible cell type, leave-one-donor-out 30-neighbor regression for total counts, detected-gene count, and mitochondrial fraction. The baseline is the same-cell-type mean over training donors. We report $R^2 = 1 - \text{NMSE}$, so higher predictability means more technical information remains accessible in the layout, not better biological quality.

3.6 Deep1B strict-ID retrieval

For public query i, let G_i^k be the first k official original-space ground-truth IDs and let L_i^k be the k nearest IDs found by exact search in the autoencoder's eight-dimensional layout. For a leading query prefix Q, we report

$$\text{Recall@}\, k = \frac{1}{|Q|k} \sum_{i \in Q} \left| L_i^k \cap G_i^k \right|, \qquad k \in \{10, 100\}. \tag{16}$$

The search makes one bounded scan of the complete base layout, uses float64 squared Euclidean distances, and maintains bounded per-query top-k state. It is strict ID recall: alternative IDs tied in original-space distance receive no credit. This tests neighbor identity after compression, not approximate-search speed or reconstruction error. Retrieval was evaluated only for the autoencoder and therefore supplies no cross-reducer retrieval ranking.

3.7 Native autoencoder reconstruction

For the final refitted model, native reconstruction is summarized by

$$\text{NMSE}_{\text{AE}} = \frac{\sum_i \|x_i - g_\phi(f_\theta(x_i))\|_2^2}{\sum_i \|x_i - \bar{x}\|_2^2}.$$

Writing $\hat{x}_i = g_\phi(f_\theta(x_i))$, mean reconstruction cosine is

$$\text{ReconCos}_{\text{AE}} = \frac{1}{N} \sum_{i=1}^{N} \frac{\hat{x}_i^\top x_i}{\max(\|\hat{x}_i\|_2, 10^{-12})}.$$

For the row-normalized text corpora, the usual factor $\|x_i\|_2$ in reconstruction cosine is one. Reconstruction cosine is not reported for the intentionally unnormalized Tabula input, while NMSE remains defined. These are autoencoder-only diagnostics; Equation (12) is the text-corpus cross-method comparison.

3.8 Controlled intrinsic-dimension cross-check

The fixed-width benchmark asks how well a chosen d preserves structure; it does not itself estimate an intrinsic dimension. We therefore run a separate controlled check on three 600-point manifolds whose local dimensions are known: a rank-three Gaussian linear subspace, a circle, and a noise-free Swiss roll. Each is orthogonally embedded in $\mathbb{R}^{10}$, independently for seeds 0, 1, and 2. Exact Euclidean input-space neighbors are used throughout. No DiRe layout is fitted in this check: the connection to DiRe is that the same neighbor-distance calculation can be applied to the input-space k-NN graph that DiRe has already constructed.

For TwoNN [8], let r_{i1} and r_{i2} be the first two neighbor radii and $\mu_i = r_{i2}/r_{i1}$. If F is the empirical CDF of the valid sorted ratios, the ideal local model obeys

$$-\log(1 - F(\mu)) = d \log \mu. \tag{17}$$

Writing $x_j = \log \mu_j$ and $y_j = -\log(1 - F_j)$, our estimate is the through-origin least-squares slope

$$\widehat{d}_{\text{TwoNN}} = \frac{\sum_{F_j < 0.9} x_j y_j}{\sum_{F_j < 0.9} x_j^2}. \tag{18}$$

Thus the noisiest upper 10% of the ratio CDF is excluded. As a second local estimate, the Levina–Bickel MLE with the MacKay–Ghahramani inverse averaging correction [9] uses $k = 20$:

$$\widehat{d}_{\text{MLE}} = \left[\frac{1}{N} \sum_{i=1}^{N} \frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{T_{ik}}{T_{ij}} \right]^{-1}. \tag{19}$$

All 600 rows in every dataset–seed cell pass the positive-distance and nondegenerate- ratio checks.

The reconstruction estimate uses a separate symmetric $10 \to 128 \to 64 \to q \to 64 \to 128 \to 10$ autoencoder for each $q \in \{1, \ldots, 6\}$. Hidden layers use SiLU and the bottleneck and output are linear. After centering by the training mean and dividing all features by one global training RMS, an 80/20 split supplies training and validation rows. AdamW uses learning rate 10^{-3}, weight decay 10^{-5}, batch size 256, a 100-epoch cap, and patience 12. Within a seed, every width uses the same split. If $\widehat{X}_{q,t}$ is the validation prediction after epoch t, the recorded normalized validation error is

$$e_q = \min_t \frac{\|X_{\text{val}} - \widehat{X}_{q,t}\|_F^2}{\|X_{\text{val}} - \bar{X}_{\text{train}}\|_F^2}. \tag{20}$$

The validation rows therefore support both early stopping and the model-selection curve; they are not an untouched test set. Each seed contributes one fit per width. The six errors are projected onto a non-increasing sequence $\widetilde{e}_q$ by isotonic regression. With

$$u_q = \frac{q-1}{5}, \qquad v_q = \frac{\widetilde{e}_q - \widetilde{e}_6}{\widetilde{e}_1 - \widetilde{e}_6},$$

the chord-elbow score is $h_q = (1 - u_q) - v_q$, and only its largest interior value is a candidate. It is accepted when $\max h_q \geq 0.05$, the relative error drop is at least 0.05, and the decrease from $q = 5$ to $q = 6$ is at most 0.01 of the total decrease. These guards prevent a flat curve or a curve still falling at the grid boundary from being reported as a dimension.

TwoNN and MLE estimate local dimension from neighbor-volume growth. The autoencoder elbow instead estimates a global, architecture- and optimization-dependent latent width for reconstruction. Their numerical comparison is therefore a cross-check between related questions, not an assertion that they are interchangeable estimators.

3.9 Sampled topology descriptions

Three label-stratified 4,000-paper arXiv samples and three uniform 3,000-paragraph Wikipedia samples are used. For each input or layout cloud Z, define its exact sampled diameter $D_Z = \max_{i,j} \|z_i - z_j\|_2$ and rescale $Z' = Z/D_Z$. Ripser persistence diagrams [11] through dimension one define Betti curves

$$\beta_q^Z(t) = \#\{(b, d) \in \mathcal{D}_q(Z') : b \leq t < d\}, \qquad q \in \{0, 1\}. \tag{21}$$

Each cloud is sampled at 50 filtration values extending to 1.05 times that cloud's own longest finite death. The reported quantity is the unnormalized FastDTW distance [12], using default radius one and local cost $|u - v|$, between the two 50-value β_q sequences. The filtration axes themselves are not supplied to FastDTW and generally differ between input and layout.

Consequently, Betti-curve DTW is a same-corpus, sampled descriptive discrepancy—not a proof of homology preservation. The nine values per method cross three reducer seeds with three evaluation samples and are not nine independent replicates. For finite positive H_0 persistences $p_1, \ldots, p_m$, supplementary "islandness" descriptions are the longest bar $\max_j p_j$, the mean of the five largest bars divided by the median, and the Gini coefficient

$$G(p) = \frac{\sum_{i=1}^m \sum_{j=1}^m |p_i - p_j|}{2m \sum_{j=1}^m p_j}.$$

These are descriptions, not distances to the input, and have no universal better direction.

For Deep1B, one already-fitted 10-million-row layout per method is uniformly reduced to a common 20,000-row prepared set. Three shared uniform 5,000-row samples use evaluation seeds 17, 29, and 43 and otherwise follow Equation (21) and the same 50-step FastDTW protocol. The samples overlap by approximately 24–26%, so their standard deviations are descriptive rather than independent uncertainty estimates. They are evaluation samples from one reducer fit per method, not three reducer fits. The 100-million-row UMAP cluster-count control is evaluated analogously on a separate common 20,000-row prepared set. Deep1B has no class labels in this evaluation.

4 Results

Boldface in the following tables marks only the best observed three-seed mean; it is not a significance claim. With three seeds, the main rankings are generally stable but uncertainty estimates are necessarily descriptive.

4.1 arXiv: $723{,}457 \times 384 \to \{2, 8\}$

d	Method	Fit (s)↓	Trust↑	Cont.↑	Overlap↑	ρ_S ↑	Stress↓	Decode↓	Accuracy↑	Bal. acc.↑
2	Autoencoder	24.32	**.877**	**.918**	.099	**.588**	.523	.647	.363	.139
2	DiRe	**15.33**	.792	.860	.116	.313	.426	.721	.377	.146
2	UMAP	30.03	.858	.902	**.147**	.489	**.418**	**.643**	**.482**	**.245**
8	Autoencoder	23.42	**.969**	**.988**	**.321**	**.801**	**.254**	**.466**	.509	.250
8	DiRe	**15.08**	.905	.897	.187	.355	.335	.534	.531	.254
8	UMAP	30.66	.937	.954	.247	.590	.357	.499	**.609**	**.329**

Table 3: arXiv means over three reducer/training seeds at both target widths. Boldface is assigned within each width. "Decode" is the shared local-decoding NMSE, not native autoencoder reconstruction.

At width two, the autoencoder is strongest on rank-aware local fidelity and pair-distance ordering, UMAP retains the most exact neighbors and predicts primary category best, and UMAP and DiRe give the two lowest stresses. At width eight, the autoencoder leads all six geometric and decoding columns, while UMAP remains clearly best on ordinary and balanced category accuracy. Increasing width from two to eight raises the autoencoder's neighbor overlap from .099 to .321 and lowers its decoding NMSE from .647 to .466; the corresponding category accuracy rises from .363 to .509 but remains below UMAP's .609. This is a within-corpus width contrast, although two tested widths do not establish a general scaling law. Figure 1 shows the seed-0 native width-two layouts and secondary PCA views of width eight, but the classifier scores in Table 3 are the quantitative label test.

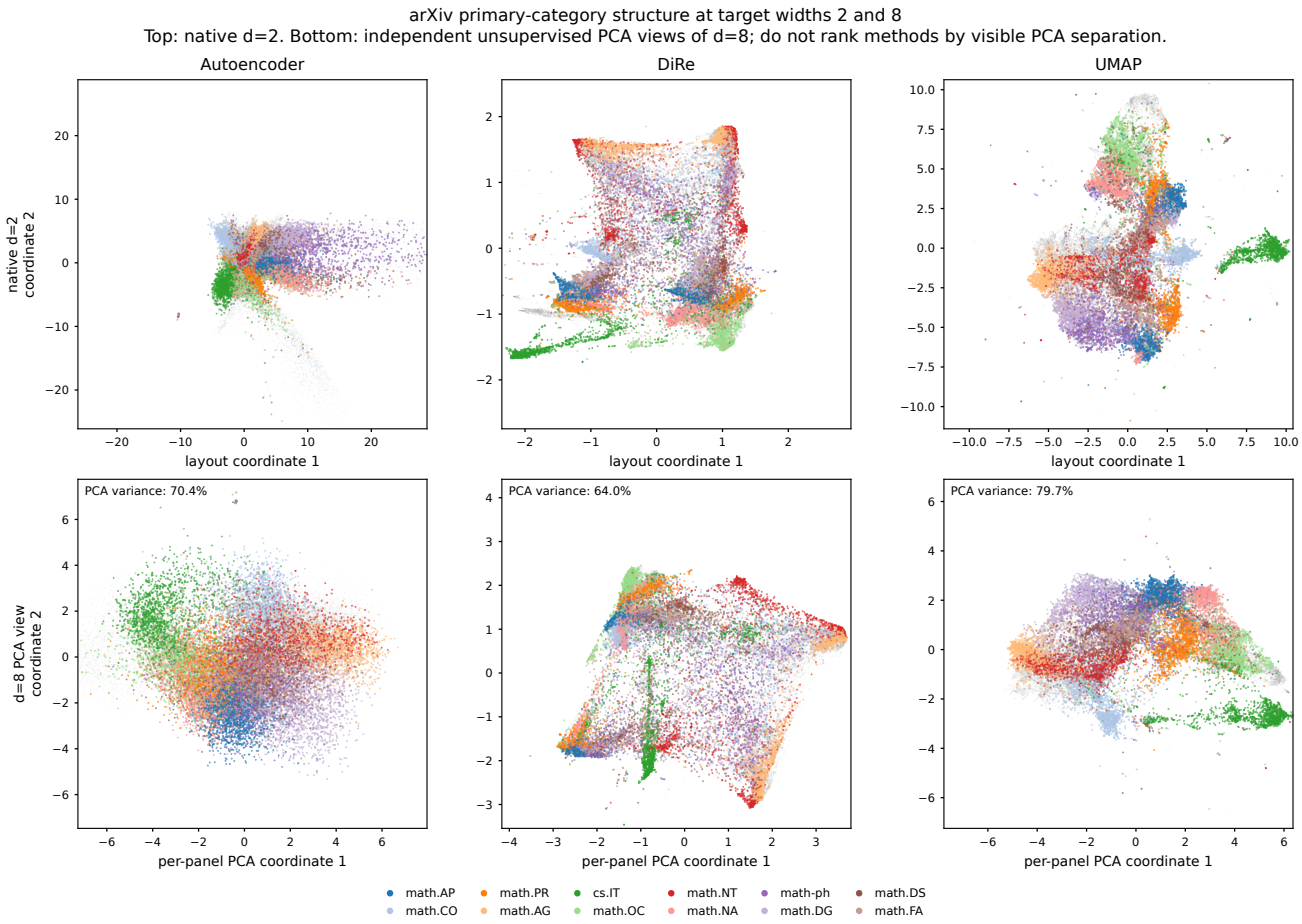


Figure 1: Seed-0 arXiv layouts at both target widths. The top row is native $d = 2$; the bottom row is an independent unsupervised PCA view of each native $d = 8$ layout, fitted on the same uniform 100,000 rows. Every panel uses 1,500 papers from each of the 12 largest primary categories and a uniform 30,000-paper background outside those categories. Colors and displayed indices are identical across widths and methods. Coordinates and isotropic robust limits are panel-specific; visible separation in the PCA row is not a cross-method score.

4.2 Wikipedia: $10^6 \times 1024 \to \{2, 8\}$

d	Method	Fit (s)↓	Trust↑	Cont.↑	Overlap↑	ρ_S ↑	Stress↓	Decode↓	Purity↑	Hit↑	TrRecall↑
2	Autoencoder	81.60	**.887**	.827	**.183**	.230	.747	.907	.006	.050	.006
2	DiRe	**62.14**	.752	.823	.118	.305	**.428**	.959	.136	.288	.141
2	UMAP	144.01	.847	**.832**	.182	**.337**	.475	**.907**	**.201**	**.498**	**.236**
8	Autoencoder	81.85	**.960**	**.944**	**.337**	**.459**	.362	**.791**	.071	.283	.079
8	DiRe	**63.93**	.908	.894	.229	.378	**.268**	.844	.306	.570	.345
8	UMAP	144.05	.917	.885	.235	.389	.438	.841	**.410**	**.754**	**.530**

Table 4: Wikipedia means over three reducer/training seeds at both target widths. Boldface is assigned within each width. Article purity, hit rate, and truncated recall query 3,000 anchors against the complete one-million-row layout. At width two, UMAP's mean decoding NMSE is only 0.00003 below the autoencoder's before rounding.

At width two, the autoencoder gives the highest trustworthiness and a narrowly higher exact-neighbor overlap than UMAP; UMAP leads continuity, pair-distance ordering, the shared decoding probe, and all three article-group scores; and DiRe gives the lowest stress. At width eight, the autoencoder wins every sample-conditional local-geometry metric and the shared decoding probe. It does not organize exact article groups as strongly: UMAP places at least one same-article paragraph among the first 16 neighbors for 75.4% of eligible anchors, compared with 57.0% for DiRe and 28.3% for the autoencoder. DiRe gives the lowest global stress at both widths. Wikipedia thus has meaningful group labels, but they express exact article identity rather than broad semantic categories. The winning method for every Table 4 width-eight metric is unchanged across the three seeds; the largest listed sample standard deviation is 0.018 for autoencoder stress.

All three reducers improve every metric in Table 4 when moving from two to eight coordinates. The autoencoder changes most sharply: stress falls from .747 to .362, pair correlation doubles from .230 to .459, and article hit rate rises from .050 to .283. UMAP remains the article-group leader at both widths, reaching .754 hit rate at width eight. The crossover therefore separates a target-width effect from the earlier corpus comparison: the autoencoder's width-eight local-geometry advantage is not merely a peculiarity of using Wikipedia rather than arXiv.

Figure 2 shows the native width-two layouts and illustrative PCA views of width eight. The PCA panels' axes and visible separations cannot be treated as the scored embedding. Figure 3 shows the native-layout retrieval statistics over both widths and all three reducer seeds.

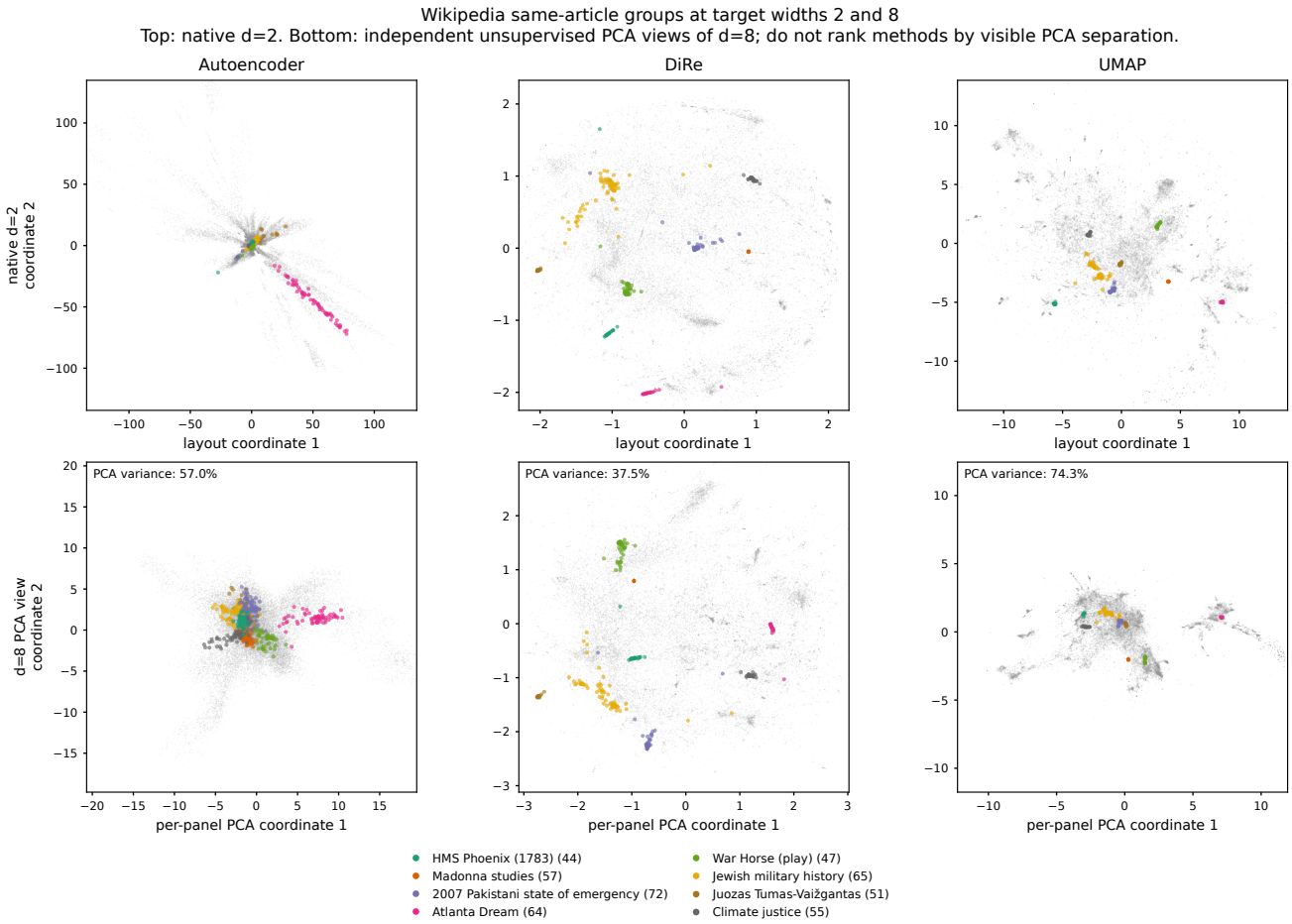


Figure 2: Seed-0 Wikipedia layouts at both target widths. The top row is native $d = 2$; the bottom row is an independent unsupervised PCA view of each native $d = 8$ layout, fitted on the same uniform 100,000 rows. Eight articles containing 40–100 paragraphs are chosen by a deterministic prespecified rule independent of all layouts, and every paragraph from them is shown over the same 30,000-row background. Highlighted and background indices are identical across widths and methods. Coordinates and isotropic robust limits are panel-specific; displayed PCA variance is descriptive, and no cross-method ranking should be inferred from the PCA row.

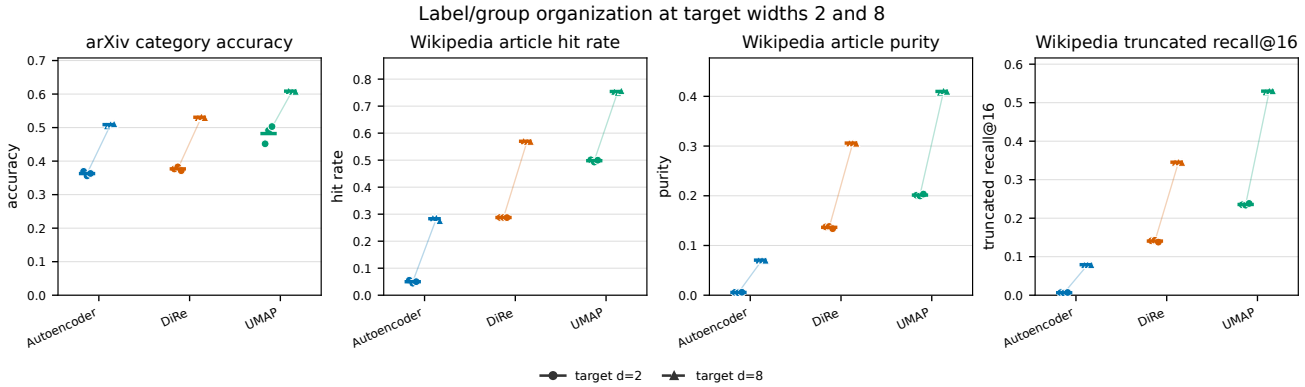


Figure 3: Label/group organization in the native layouts. Points are the three reducer/training seeds and horizontal segments are means; circles and solid mean bars are $d = 2$, triangles and dashed bars are $d = 8$, and faint segments join width means. arXiv uses category classification accuracy. Wikipedia uses full-million-layout article hit rate, purity, and truncated recall at 16 with the same 3,000 anchors at both widths. These are native-layout metrics, not scores of the PCA views in Figure 2. Labels are evaluation-only and are not supplied to any reducer.

4.3 Tabula Sapiens: $100{,}000 \times 50 \to \{2, 8\}$

d	Method	Fit (s)↓	Trust↑	Cont.↑	Overlap↑	ρ_S ↑	Stress↓	Macro-F1↑	Bal. acc.↑	ASW↑
2	PCA	**.12**	.767	.921	.110	.441	.441	.052	.057	-.393
2	Autoencoder	27.05	.952	.969	.388	**.723**	.448	.233	.244	-.330
2	DiRe	3.66	.935	.957	.451	.259	.470	.285	.291	-.181
2	UMAP	.94	**.964**	**.972**	**.491**	.370	**.434**	**.331**	**.335**	**-.127**
8	PCA	**.12**	.954	.985	.395	.800	**.211**	.180	.189	-.144
8	Autoencoder	27.79	**.988**	**.985**	**.565**	**.843**	.255	.267	.277	-.113
8	DiRe	3.88	.972	.978	.522	.415	.316	.307	.317	-.081
8	UMAP	1.36	.979	.980	.553	.408	.334	**.349**	**.353**	**-.074**

Table 5: Tabula Sapiens PCA50 means over three reducer/training seeds. Geometry is relative to the complete prepared PCA50 representation, with neighborhood scores on the fixed 5,000-cell sample. Cell-type classification uses donor-held-out folds; ASW uses the separate fixed 5,000-cell stratified sample. Fit timing is reducer-specific: autoencoder fit includes validation selection, fresh full-data refit, encoding, and reconstruction, while graph-method fit is synchronized `fit_transform`. Boldface marks the best observed mean within each width, not significance.

At width two, UMAP leads four of the five geometric scores and all three biological scores, while the autoencoder gives the highest pair-distance rank correlation. At width eight, the autoencoder leads trustworthiness, continuity, exact overlap, and pair correlation; PCA gives the lowest stress, and UMAP remains strongest on every cell-type score. The autoencoder's native reconstruction NMSE falls from .1523 at width two to .0541 at width eight. Thus good reconstruction and PCA50-relative geometry do not imply the strongest donor-held-out biological organization. All ASWs remain negative, including UMAP's -.074 at width eight, so none of these layouts yields uniformly compact global cell-type clusters.

Detected-gene leakage supplies the opposite diagnostic. At width two its cell-type-conditioned R^2 is .279 for DiRe, .292 for UMAP, .400 for the autoencoder, and .567 for PCA; at width eight the values are .311, .359, .380, and .543, respectively. Lower is preferable here. DiRe therefore retains the least predictable detected-gene-count signal in this probe even though UMAP retains cell type more strongly. No single score should collapse those two objectives.

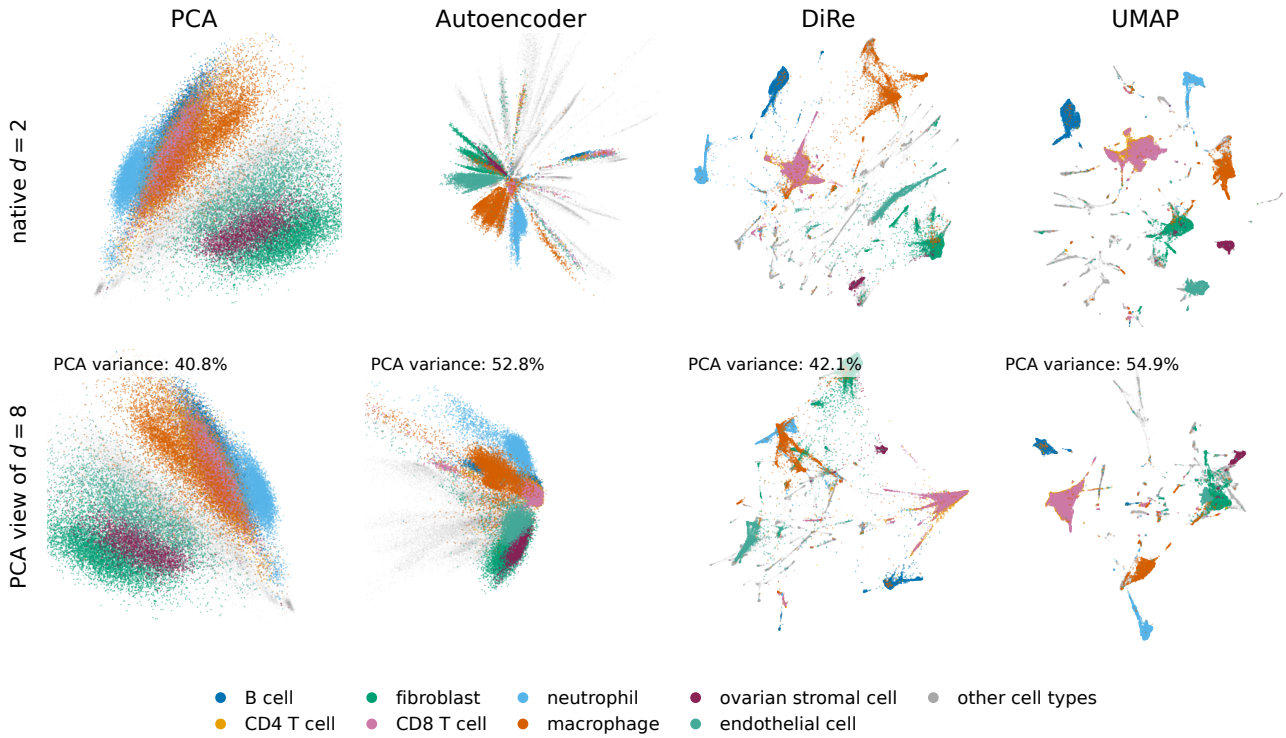


Figure 4: Seed-0 Tabula Sapiens cell-type organization. The top row contains native $d = 2$ layouts; the bottom row is a separate unsupervised PCA view of each native $d = 8$ layout, with the retained two-component variance printed in-panel. All 100,000 cells are displayed. The eight most frequent of 180 annotated cell types use consistent colors, and the other 172 types are gray; labels are never supplied to a reducer. Coordinates and isotropic 99.5% robust limits are panel-specific. The bottom row is a descriptive view, not a cross-method score; Table 5 contains the native eight-dimensional metrics.

The higher working degree changes graph recall, not the conclusion. Table 6 isolates the NN-descent stage. Moving from DiRe's default 31/62 working degrees to the UMAP-like 64/128 setting raises mean Recall@30 by .000156 (0.0156 percentage points) and the fraction of completely exact 30-neighbor rows by .0044. Recall@1 and Recall@10 do not improve. Mean graph time rises by 3.3%, from .255 to .264 seconds, as expected from the larger working graph.

Working degrees	Recall@1↑	Recall@10↑	Recall@30↑	Exact rows@30↑	Graph (s)↓
31/62 (DiRe default)	.984600	.993320	.995851	.8825	.255
64/128 (UMAP-like)	.984533	.993313	.996007	.8869	.264

Table 6: Means over three independent approximate-graph builds at fixed final $k = 30$. Recall and exact-row fractions use the same 5,000 exact brute-force queries against all 100,000 cells. The Recall@30 sample standard deviations are 7.1×10^{-5} and 4.1×10^{-5}; all three builds satisfy the no-self, no-duplicate, and in-range graph contract.

Reusing each audited graph for paired layouts produces no material quality shift (Figure 5). Across the eight plotted geometry and cell-type metrics, the largest absolute change in a three-seed mean is .001995: width-two balanced accuracy slightly decreases. The individual paired label-probe changes are noisier than the geometry changes, which is why the figure exposes all three seeds. With the larger degrees, width-two DiRe still has trustworthiness .936, overlap .452, and macro-F1 .283, versus UMAP's .964, .491, and .331; the corresponding width-eight values are .972/.522/.307 versus .979/.553/.349. DiRe nevertheless retains better width-eight pair correlation and stress than UMAP. The controlled result therefore points to downstream objective/layout differences, not a low-recall DiRe graph, as the main source of the remaining local and cell-type gap in this cell.

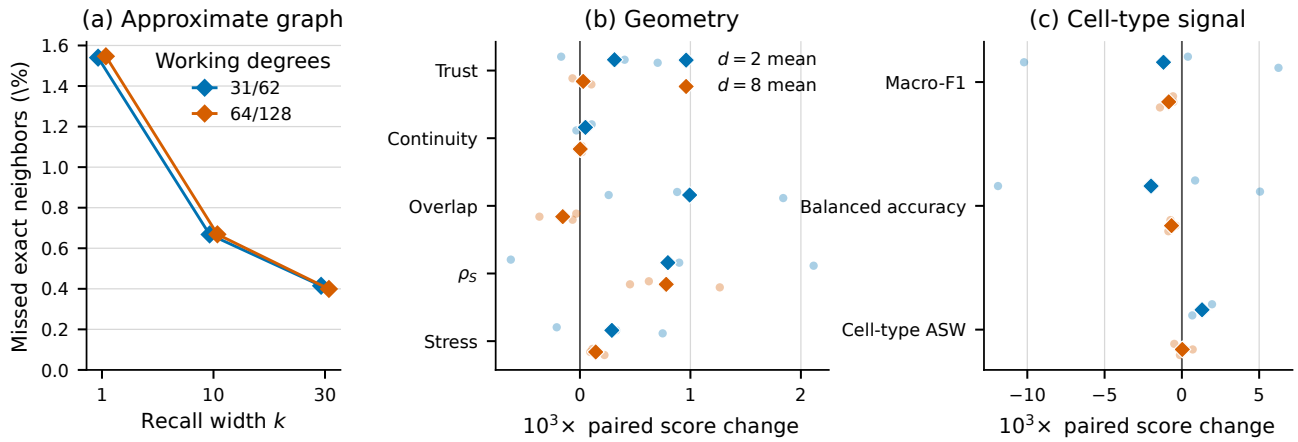


Figure 5: Controlled DiRe NN-descent working-degree audit. Panel (a) reports missed exact neighbors; small translucent points are three independent graph builds and diamonds joined by lines are means. Panels (b)–(c) report paired 64/128-minus-31/62 downstream changes after reusing each graph for both target widths; stress is sign-reversed so positive always means improvement. Small points are the three seedwise changes and diamonds are means. The horizontal scale is expanded by 10^3. "UMAP-like" refers only to degree settings inside DiRe's RAPIDS 26.06 all-neighbors builder; no UMAP graph is extracted.

4.4 Sampled topology descriptions

Corpus	d	Method	FastDTW(β_0)↓	FastDTW(β_1)↓
arXiv	2	Autoencoder	4,841	439
	2	DiRe	**3,322**	**176**
	2	UMAP	10,414	941
	8	Autoencoder	**1,583**	3,045
	8	DiRe	2,685	**177**
	8	UMAP	6,469	499
Wikipedia	2	Autoencoder	4,822	**8,974**
	2	DiRe	**2,107**	9,355
	2	UMAP	3,566	9,207
	8	Autoencoder	**1,591**	**7,572**
	8	DiRe	3,335	9,117
	8	UMAP	2,986	9,198

Table 7: Pooled descriptive means across three reducer seeds and three evaluation samples. Lower raw sequence discrepancy is preferable within a corpus and target width, subject to the sampling, cloud-specific filtration grids, and FastDTW qualifications in the text.

At arXiv width two, DiRe has the lowest sampled β_0 and β_1 curve discrepancies, and the autoencoder lies between DiRe and UMAP. At width eight the result splits: the autoencoder has the lowest β_0 discrepancy, while DiRe remains lowest on β_1. Increasing arXiv width lowers β_0 discrepancy for all three methods, but β_1 is nearly unchanged for DiRe, improves for UMAP, and worsens substantially for the autoencoder. Wikipedia behaves differently. At width two, DiRe has the lowest β_0 discrepancy and the autoencoder the lowest β_1 discrepancy; at width eight the autoencoder has the lowest value in both columns. From width two to eight, both autoencoder discrepancies fall, whereas DiRe's β_0 discrepancy rises and UMAP's β_1 discrepancy is essentially unchanged. Thus neither a wider target nor one reducer has a corpus-independent topological advantage in these samples; the rankings exhibit a corpus-by-width interaction. Table 7 is a descriptive sample comparison, not a claim that the reported values prove preservation of the full-corpus homology.

4.5 Local and reconstructive dimension estimates are related, not identical

Manifold	Known local d	TwoNN	MLE ($k = 20$)	AE elbow
Rank-three linear	3	3.036 ± 0.094	3.053 ± 0.045	3.000 ± 0.000
Circle	1	1.034 ± 0.061	0.997 ± 0.008	2.000 ± 0.000
Swiss roll	2	1.990 ± 0.084	1.929 ± 0.033	2.667 ± 0.577

Table 8: Controlled input-space intrinsic-dimension cross-check. Entries are mean±sample standard deviation across three independent data/model seeds. The AE elbows for the Swiss roll are 3, 3, and 2. These synthetic results are not estimates for the arXiv or Wikipedia corpora.

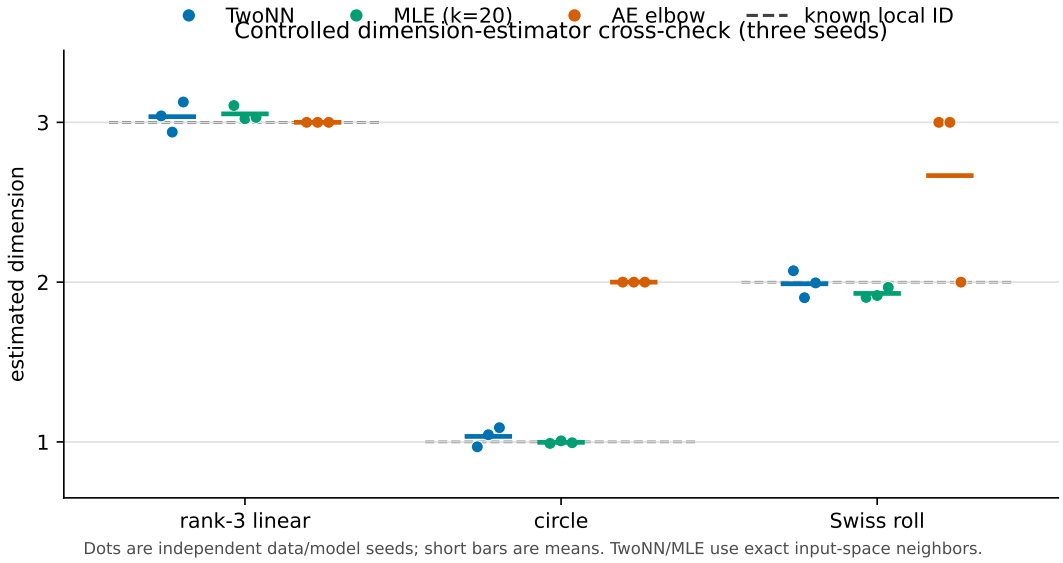


Figure 6: The three seeded estimates underlying Table 8. Dots show individual seeds, short horizontal segments show means, and dashed segments show known local dimension. TwoNN and MLE use exact input-space neighbors; the AE points are accepted validation-reconstruction elbows.

Rounding TwoNN to the nearest integer, its estimate and the AE elbow agree exactly in four of nine seeded trials and differ by at most one in all nine. Their paired Spearman correlation is $\rho_S = .779$. This correlation is descriptive only: the nine pairs are clustered within three deliberately chosen manifold families, so neither a null-model p-value nor a population-level inference is warranted.

The pattern is more informative than the aggregate count. Both estimators recover the rank-three linear subspace. The circle is locally one-dimensional, but a continuous Euclidean autoencoder generally needs two coordinates to reconstruct it globally without a cut, so its stable 1 versus 2 difference reflects the two estimands rather than a failure. A Swiss roll is locally two-dimensional and globally unrollable into a plane; TwoNN recovers two, while finite-network optimization together with the chord-elbow rule selects either two or three across seeds. Thus the pointwise and neighbor-based approaches do not produce unrelated notions of dimension, but topology, model capacity, optimization, and elbow-rule sensitivity prevent an identity theorem.

4.6 Why is the autoencoder so fast?

Corpus	d	GPU staging	Epoch selection	Full refit	Encode/statistics	Total
arXiv	2	0.14	11.90	11.58	0.69	24.32
arXiv	8	0.11	11.43	11.21	0.67	23.42
Wikipedia	2	0.40	39.14	39.60	2.46	81.60
Wikipedia	8	0.42	39.29	39.67	2.47	81.85

Table 9: Mean autoencoder fit-time decomposition in seconds. The headline time includes both training phases.

The timing is less paradoxical than it first appears. The arXiv and Wikipedia networks are small, their complete 1.1 and 4.1 GB float32 matrices fit in GPU memory, and a batch of 32,768 leaves only about 23 and 31 steps per epoch. Approximately 400 epochs across selection and refitting therefore require about 8,800 and 12,000 regular dense optimizer steps after accounting for the 5% selection holdout. The primary results in Table 9 use no mixed precision and include both the validation-selection model and the fresh full-data model.

These are implementation measurements on a GH200, not hardware-independent complexity constants. Both graph reducers use sparse $O(Nk)$ layout state, but graph construction is a separate cost. The seeded text UMAP runs use exact brute-force search, whereas DiRe and the larger UMAP arms use approximate indices; their timing therefore combines different search and layout kernels rather than one universal graph-method complexity.

4.7 Deep1B scaling: bounded autoencoder state through 100 million rows

Method/configuration	Rows	Core (s)	AE encode (s)	Total/worker (s)	RSS (GB)	Torch alloc. (GB)	NVML increment (GB)	Outcome
AE FP32	10^7	5.45 train	3.19	12.16	1.48	.150	–	NMSE .45293
AE FP32	10^8	52.86 train	31.71	88.18	2.02	.150	–	NMSE .39255
UMAP, 16 clusters	10^7	66.40 fit	–	68.82	12.13	–	7.92	complete
UMAP, 16 clusters	10^8	861.46 fit	–	874.79	104.48	–	77.77	complete
UMAP, 160 clusters	10^8	836.65 fit	–	847.41	102.07	–	77.84	complete
DiRe	10^7	158.14 fit	–	161.46	19.82	–	29.06	complete
DiRe	10^8	–	–	2694.38	211.10	–	63.03	CUDA OOM

Table 10: Single-run Deep1B GPU scaling measurements; memory values are decimal GB. "Core" means autoencoder training or graph `fit_transform`. Autoencoder total and graph worker time have the broader, method-specific scopes defined in the protocol. PyTorch allocator memory and incremental NVML process memory are different scopes and should not be treated as a common total-memory measurement. The failed DiRe allocation never became resident.

From 10 million to 100 million rows, FP32 autoencoder training and encoding take 9.71× and 9.94× as long, respectively. Application peak RSS rises from 1.48 to 2.02 GB, while PyTorch peak allocated memory is exactly 150,232,064 bytes in both runs. The unavoidable float32 layout grows from 0.32 to 3.20 GB. These measurements support approximately linear work with bounded model and minibatch state for this one-pass implementation. They do not imply literally unbounded scaling: input I/O, output storage, and every additional pass still grow with N. Held-out NMSE falls from .45293 to .39255, but the runs make 1,221 and 12,208 optimizer updates; the change therefore combines more data with more optimization and is not a controlled sample-size-only quality comparison. The controls below show that distinct-pool effects are small at 1,204 updates and that a descriptive tune-cohort separation appears under longer optimization; they do not identify a causal decomposition for the separate 100-million-row run.

UMAP also completes at 100 million rows. With 16 clusters, fit time grows by 12.97×, application RSS from 12.13 to 104.48 GB, and incremental GPU-process memory from 7.92 to 77.77 GB. Holding the cluster count fixed also lets the estimated overlap-aware partition grow from 1.25 million to 12.5 million rows. Raising the 100-million-row cluster count to 160 restores the estimated partition to 1.25 million rows and reduces its estimated input from 4.8 to 0.48 GB, but the incremental GPU peak remains 77.84 GB. This control points to global graph/layout state, rather than partition input alone, as the dominant measured peak. With one non-deterministic NN-descent fit per arm, the small timing and RSS changes do not establish a stable cluster-count effect.

The tested DiRe configuration completes at 10 million rows, but its worker fails after 2,694.38 seconds during the 100-million-row fit; the backend returns no fit time. The recorded failure is a request for a 102.4 GB (95.37 GiB) full negative-index tensor on a 94.50 GiB GPU. It is an observed limit of the tested cuVS-backed vectorized-force path with $k = 16$, negative ratio eight, and 128 iterations, not a claim that every partitioned or reformulated DiRe implementation must fail at this scale. The pilot ceiling is 100 million, not one billion: the full base file requires 384 GB and its float32 eight-dimensional layout another 32 GB, while only 218.4 GB of local filesystem space was available when the bundle was created.

(a) Exact AE-layout retrieval

Base / queries	Recall@10	Recall@100
10^7 / 100	.0300	.0595
10^7 / 1,000	.0358	.0677
10^8 / 100	.0210	.0449

(b) 10M sampled topology

Method	FastDTW(β_0)	FastDTW(β_1)
Autoencoder	1628 ± 219	1954 ± 91
DiRe	4167 ± 304	10940 ± 292
UMAP	8943 ± 1138	13528 ± 379

Table 11: Deep1B quality diagnostics. Panel (a) is AE-only strict ID recall under Equation (16); 10 million uses a CPU-trained model and 100 million a GPU-trained model. All rows use leading public-query prefixes, with query count as shown. Panel (b) gives means±sample standard deviations across three shared 5,000-point evaluation samples from one fitted 10-million-row layout per method; its autoencoder layout is also the CPU-trained arm. Lower FastDTW is preferable subject to the sampled-topology qualifications in the text.

The retrieval result is deliberately sobering. Even the larger 10-million-row query arm recovers only 3.58% of the first ten official IDs and 6.77% of the first hundred. The apparent decrease at 100 million is not a controlled row-count effect: base and ground truth change, query counts differ, and the two layouts use CPU- and GPU-trained models. More importantly, strict neighbor identity and reconstruction MSE ask different questions. Fast bounded-state training does not make their answers equivalent.

The autoencoder has the smallest sampled Betti-curve discrepancy in both 10-million-row columns, but finite-H_0 descriptions are mixed: it is closest to the reference in top-five-to-median ratio and Gini coefficient, while UMAP is closest in longest finite H_0 bar. With one fitted layout per method and overlapping evaluation samples, these are descriptive differences rather than a definitive topology ranking. At 100 million rows, the UMAP 16- and 160-cluster controls are effectively tied: their (β_0, β_1) FastDTW means are $(10334.0, 13167.7)$ and $(10312.7, 13193.7)$. Paired changes have mixed signs and average only -0.206% and $+0.197\%$, giving no evidence here of a material cluster-count topology effect.

The 10-million-row BF16 arm supplies one further small-network caution. BF16 slows training by 15.7% (6.30 versus 5.45 seconds) and total time by 7.7%, while held-out NMSE changes from .45293 to .45339 and PyTorch peak allocated memory falls from 150.2 to 122.5 MB. BF16 ran immediately before FP32, so cache and order effects are not controlled; this supports shape-dependent overhead, not an intrinsic BF16 slowdown.

4.8 Deep1B controls: distinct rows versus optimizer exposure

The one-pass quality curve changes both the number of distinct examples and the number of optimizer updates. Table 12 separates those variables on the fixed 10-million-row universe. Its one-pass tune column reproduces the steep apparent sample-size response: NMSE falls from 1.01715 after eight updates to .45507 after 1,204. When all arms instead receive 1,204 updates, tune and legacy-evaluation reconstruction become nearly flat across five nested pools.

Distinct pool	One-pass updates	Effective passes	One-pass tune NMSE	Fixed-budget tune NMSE	Selected legacy NMSE
65,536	8	150.500	$1.01715 \pm .00195$	$.45586 \pm .00200$	$.45514 \pm .00177$
262,144	32	37.625	$.99500 \pm .00525$	$.45480 \pm .00167$	$.45384 \pm .00161$
1,048,576	128	9.406	$.81228 \pm .01432$	$.45450 \pm .00219$	$.45366 \pm .00203$
4,194,304	512	2.352	$.51782 \pm .00080$	$.45466 \pm .00227$	$.45381 \pm .00208$
9,863,168	1,204	1.000	$.45507 \pm .00235$	$.45507 \pm .00235$	$.45422 \pm .00227$

Table 12: Deep1B training-pool control; entries are mean±sample standard deviation over three paired initialization seeds. Every fixed-budget arm sees exactly 9,863,168 row presentations. The tune columns share one cohort; legacy evaluation occurs only after tune-only checkpoint selection. The one-pass column deliberately varies both distinct rows and update count.

At the common budget, the five legacy NMSE means span only .45366–.45514. The 65,536-row value is .000918, or 0.20%, above the 9,863,168-row value. Endpoint geometry changes are also small: trustworthiness .96160 to .96292, continuity .97062 to .97107, overlap .31026 to .31143, pair-distance Spearman .49548 to .50434, stress .27682 to .27462, and local-decoder NMSE .51990 to .51959. These are descriptive three-seed differences, generally comparable to initialization variation rather than evidence of formal equivalence. The seed-1729 retrieval spot check is likewise nearly flat: Recall@10 is .0343 versus .0349 and Recall@100 is .0656 versus .0679 for the smallest and largest pools.

The fixed 1,204-update result does not establish that 65,536 rows or 1,204 updates are universally sufficient. Figure 7 extends the primary 65,536-row pool and a 1,048,576-row robustness pool through 9,632 uninterrupted updates. Discovery is tune-only, and every update-1,204 state exactly reproduces the preceding experiment.

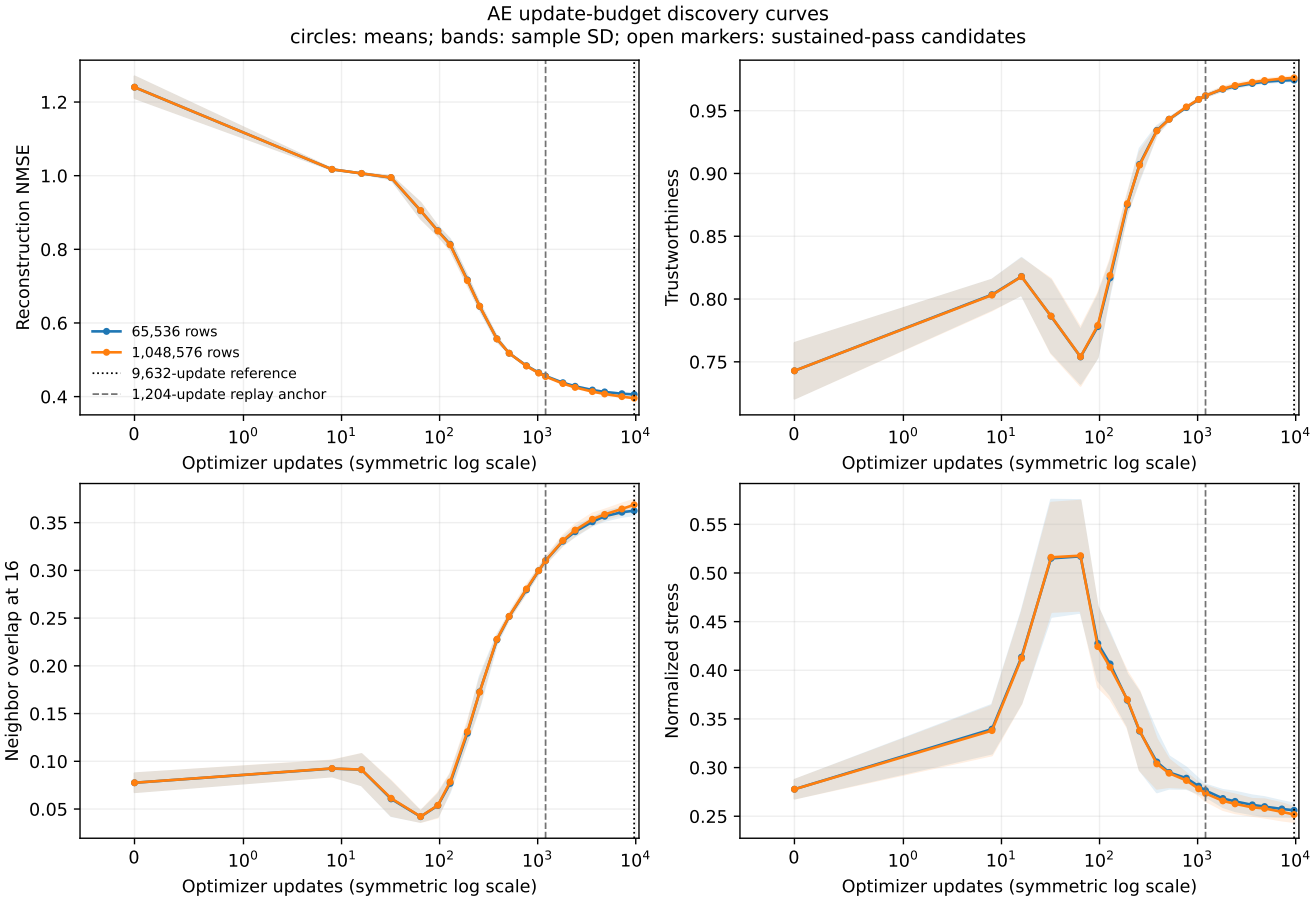


Figure 7: Deep1B update-budget discovery. Curves are means and shaded bands are sample standard deviations over three paired seeds. The dashed line marks the exact 1,204-update replay anchor and the dotted line the tested 9,632-update terminal reference. Reconstruction and geometry use only the tune cohort and its fixed samples. No checkpoint satisfies the prespecified sustained-pass gate; the terminal reference is not an established optimum.

Relative to 9,632 updates, 1,204-update tune NMSE is 12.3% worse for the 65,536-row pool and 14.9% worse for the one-million-row pool. Trustworthiness and exact overlap rise while stress and local-decoder NMSE fall over the same trajectories. Optimizer time grows from about 4.5 to 35.3 seconds per trajectory on the GH200. The final 7,224-to-9,632 NMSE changes remain .462% and 1.077%, both above the frozen .25% flatness threshold. Thus no qualifying plateau under the frozen tune-NMSE rule and no stopping candidate are identified through the tested cap; neither fresh-seed legacy confirmation nor long-budget retrieval is run.

Longer optimization also yields a descriptive tune-cohort pool-by-budget separation. At 1,204 updates the two pools are nearly tied, whereas at 9,632 the one-million-row pool has lower NMSE (.39564 versus .40596), higher overlap (.36872 versus .36244), and lower stress (.25176 versus .25611). The defensible conclusion is therefore conditional: observed differences between pools are small at the short common budget in this fixed draw, while the larger pool is associated with better tune metrics under longer optimization. Whether that separation repeats on the legacy cohort and fresh seeds is untested. One-pass throughput, sample efficiency, and time to matched quality are different estimands.

4.9 BF16 sensitivity: tested margins pass, topology is mixed, and speed is model-size dependent

We repeated every autoencoder cell—both corpora, $d \in \{2, 8\}$, and seeds 0–2—with CUDA BF16 autocast during training, epoch-selection validation, and final encoding. Parameters, gradients, AdamW state, the explicit MSE loss, and reconstruction accumulations remained float32. Inputs, samples, seeds, architecture, and optimizer settings matched the primary runs. These are paired sensitivity runs; the scientific comparison tables remain the primary float32 results.

		Selection + refit training			Complete reported fit			Native NMSE
Corpus	d	FP32 (s)	BF16 (s)	Speedup	FP32 (s)	BF16 (s)	Speedup	BF16/FP32
arXiv	2	23.49	25.49	0.920×	24.32	26.28	0.925×	1.00193
arXiv	8	22.64	25.46	0.889×	23.42	26.25	0.892×	1.00131
Wikipedia	2	78.74	39.43	1.997×	81.60	42.28	1.931×	1.00096
Wikipedia	8	78.96	41.53	1.901×	81.85	44.34	1.846×	1.00023

Table 13: Paired three-seed BF16 sensitivity. Times are arithmetic means. Speedup is FP32 time divided by BF16 time; training speedup is the paired geometric mean and total speedup is the paired arithmetic mean. The displayed NMSE ratio is the arithmetic mean of the three paired ratios. Values below one indicate a BF16 slowdown.

Both the standard and FP32-evaluation-isolation BF16 arms pass all prespecified practical checks on native reconstruction and shared geometry or label metrics in every cell; with only three seeds these margins are descriptive, not formal equivalence tests. The largest paired BF16/FP32 native-NMSE ratio in the standard arm is 1.00243. The sampled-topology response is mixed rather than uniformly invariant. Pooled (β_0, β_1) DTW means change from $(4841, 439)$ to $(5660, 537)$ for arXiv $d = 2$, from $(1583, 3045)$ to $(1659, 3068)$ for arXiv $d = 8$, from $(4822, 8974)$ to $(4820, 8986)$ for Wikipedia $d = 2$, and from $(1591, 7572)$ to $(1505, 8121)$ for Wikipedia $d = 8$. A clean rerun that trained with BF16 but performed validation, encoding, reconstruction, shared-metric evaluation, and topology input generation in float32 left shared-metric quality essentially unchanged and moved the pooled topology mean closer to FP32 in only four of eight corpus–width–Betti comparisons. Thus this descriptive isolation does not support final-coordinate quantization as the main explanation for the topology differences.

A 100-step resident-batch microprofile explains why a universal three- or four-fold speedup does not appear. With TF32 matrix multiplication disabled, this fixed-order profile uses a synthetic standard-normal resident batch and separate warm-ups for each precision. Its FP32/BF16 total-step speedups are 0.859 and 0.842 for the two arXiv widths, versus 2.407 and 2.428 for Wikipedia; the Wikipedia forward-plus-backward portion alone reaches about 2.65×. The approximately 272,000-parameter arXiv MLP is overhead-bound at this batch shape, so autocast dispatch and conversions outweigh faster eligible kernels. The roughly 1.35-million-parameter Wikipedia MLP supplies enough dense matrix work to benefit, although FP32 AdamW and the non-training parts of the fit cap the end-to-end gain near 1.85–1.93×. Because the full-fit FP32 runs and later BF16 runs were not alternated contemporaneously, their timing ratios are descriptive; the microprofile independently reproduces the workload crossover.

4.10 Longer MSE optimization is not a geometry rule

The Deep1B sweep asks whether a shorter run reproduces a longer MSE-trained reference on monitored reconstruction and local-geometry diagnostics. The following sensitivity check asks a different question: whether further MSE optimization improves every notion of a useful layout.

Corpus	Run	Native NMSE↓	Trust↑	Cont.↑	Overlap↑	ρ_S ↑	Stress↓	DTW β_0 ↓	DTW β_1 ↓
arXiv	primary (199 epochs)	.5965	.8818	.9140	.1006	.5925	.5615	4,544	516
arXiv	longer (398 epochs)	.5753	.8940	.9091	.1135	.5746	.6955	5,771	772
Wikipedia	primary (200 epochs)	.6910	.9600	.9426	.3339	.4593	.3818	1,599	8,146
Wikipedia	longer (400 epochs)	.6744	.9629	.9396	.3460	.4517	.3969	1,545	9,709

Table 14: Reducer-seed-0 400-epoch-cap sensitivity checks for the two original benchmark cells only: arXiv $d = 2$ and Wikipedia $d = 8$. DTW columns average the three fixed evaluation samples. Longer training lowers native reconstruction error and improves some local scores, while several global and sampled-topology scores worsen.

For these two original cells, lower reconstruction error is therefore not a universal indicator of a more useful layout. Longer training improves native MSE, trustworthiness, and exact overlap, but reduces continuity and pair-rank correlation, worsens stress, and gives mixed or worse sampled Betti-curve discrepancies. No analogous 400-epoch claim is made for the crossed target widths. A geometry-aware autoencoder objective might change this tradeoff, but it would also move away from the deliberately simple pointwise MSE experiment studied here.

5 Conclusions and limitations

The reducer results support eight narrow conclusions.

1. A small MSE autoencoder is computationally practical at 723,000–1,000,000 rows in this GH200 configuration. Its favorable timing is consistent with fixed-architecture linear-in-N pointwise work and efficient dense kernels. BF16 autocast passes the tested reconstruction and shared-metric practical margins, while its sampled-topology response is mixed. Its speed is shape-dependent: it nearly halves Wikipedia training time and slightly slows the smaller arXiv model.
2. The separate Deep1B pilot supports bounded-state, approximately linear one-pass autoencoder processing through 100 million rows. UMAP also completes at that scale, but with graph/layout memory that grows strongly with N; the tested DiRe path reaches a configuration-specific CUDA OOM. At the common 1,204-update budget, five nested training pools have legacy NMSE means between .45366 and .45514; the steep tune-cohort one-pass curve therefore cannot be interpreted as a sample-size-only effect. This is not a universal 65,536-row sufficiency result: no qualifying plateau is observed under the frozen tune-NMSE rule through 9,632 updates, and the one-million-row pool has descriptively better tune metrics than 65,536 rows under longer optimization, without fresh-seed legacy confirmation. Strict-ID recall remains low, so systems scalability and nearest-neighbor utility must be assessed independently. Storage limits prevented a one-billion-row run.
3. Plain reconstruction can preserve sampled local geometry and decodability very well without an explicit pairwise objective. This does not guarantee strong label/group organization: UMAP wins arXiv category prediction at both widths and Wikipedia same-article retrieval at both widths.
4. The Tabula Sapiens pilot extends that separation to donor-held-out biology. UMAP leads all cell-type scores at both widths, while the autoencoder leads four of five PCA50-relative geometry scores at width eight. DiRe's default graph already has .99585 Recall@30 on the exact-query sample; doubling its working degrees changes no plotted downstream mean by .0020. In this controlled cell, graph approximation does not explain the remaining UMAP–DiRe quality gap.
5. On both corpora, increasing d from two to eight changes local-fidelity rankings as well as score magnitudes: the autoencoder leads the local, distance-order, and decoding columns at width eight but not at width two. DiRe retains the best Wikipedia stress. This repeated contrast supports a target-width effect within the tested settings, not a general scaling law.
6. Controlled known-manifold checks show approximate rather than exact agreement between TwoNN local dimension and the AE reconstruction elbow. Their systematic circle difference illustrates why local and global reconstructive dimension must be interpreted separately.
7. More MSE optimization is not a universal stopping rule or geometry objective. Deep1B reconstruction and monitored local geometry improve through the tested budget, whereas the two original text-corpus sensitivity cells show that better MSE and local scores can accompany worse global-distance or sampled-topology scores.
8. Sampled topology has no universal method or width ordering here. The arXiv width-eight autoencoder improves markedly in β_0 but worsens in β_1; the Wikipedia width-eight autoencoder improves in both. These completed cells point to a corpus-by-width interaction rather than a single topological winner.

The comparison has important limits. It uses one accelerator, one architecture family, one neighborhood size, and only two target widths. arXiv category labels and Wikipedia URL groups have different meanings. All primary layouts are transductive, and the autoencoder is retrained for every corpus, width, and seed. Neighborhood metrics are sample-conditional; distance metrics use sampled pairs; topology uses small sampled clouds and cloud-specific curve grids. The three-seed comparisons are descriptive, not formal uncertainty quantification, and two target widths cannot identify a general dimension-response curve. The controlled dimension check uses small synthetic manifolds and does not estimate the intrinsic dimension of either text corpus. The Deep1B scaling arms add further asymmetries: each completed arm contributes one fitted layout, the failed DiRe arm contributes none, the autoencoder receives one pass, and UMAP NN-descent is not exactly deterministic. Local-file page cache was not controlled, and timing and memory scopes differ across implementations. Strict-ID retrieval was evaluated only for

the autoencoder and gives no alternative-ID distance-tie credit. Finally, 100 million is the tested storage-limited ceiling, not evidence from a billion-row execution.

The single-cell evidence is one 100,000-cell, transductively preprocessed, stratified pilot from one versioned Tabula Sapiens dataset. PCA50 is the only completed input arm; the 2,000-HVG sensitivity and full 1.14-million-cell cohort are not reported. Cell-type annotations are not independent experimental ground truth, cells within donors are dependent, and donor-held-out probes do not make the reducer itself inductive. The graph audit checks exact neighbors for 5% of cells, uses three descriptive builds, and matches UMAP-like degrees inside DiRe rather than comparing raw DiRe and UMAP graphs. It identifies no degree effect under these settings, not equivalence of graph builders or layout objectives.

The Deep1B controls are also narrow. The pool-size study uses one nested training-pool draw, one $96 \to 8$ architecture, one 1,204-update budget, and three initialization seeds; it does not test reconstruction-elbow stability. Long-budget discovery uses only the 65,536- and 1,048,576-row pools and the tune cohort. Because no candidate advances, there is no fresh-seed legacy or retrieval confirmation, no long-budget topology or label test, and no evidence beyond 9,632 updates. Neither 1,204 nor 9,632 is an established optimum. The practical margins are engineering choices, and optimizer timing excludes repeated geometry evaluation and remains GH200-specific. No reducer is tuned to matched quality or convergence.

Reproducibility

The public companion *dimension-reduction-benchmarks* repository [19] provides the benchmark, topology, controlled-dimension, and Deep1B scaling, training-pool, and optimizer-budget implementations together with the single-cell preparation, fixed-width comparison, DiRe graph audit, compact result tables, and machine-readable provenance. The controlled-study supplements include public-safe CSV summaries, the frozen update-budget protocol, replay verification, and PDF figures. Raw Deep1B arrays, checkpoints, and generated layouts are not included. The supplementary records identify the official sources, prefix derivations, environment, configurations, timing scopes, checksums, sample identities, failure state, and topology summaries; the text-corpus supplements also retain complete topology curves. Those audit identifiers are intentionally kept out of the public-facing narrative. The compact Deep1B bundle is the record of the reported runs and pins the historical run-harness identities; live convenience harnesses may evolve independently.

References

[1] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science* 313 (2006), 504–507, https://doi.org/10.1126/science.1127647.

[2] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction," arXiv:1802.03426 (2018), https://doi.org/10.48550/arXiv.1802.03426.

[3] A. Kolpakov and I. Rivin, "Dimensionality reduction for homological stability and global structure preservation," arXiv:2503.03156 (2025), https://doi.org/10.48550/arXiv.2503.03156.

[4] A. Kolpakov and I. Rivin, *dire-rapids: PyTorch and RAPIDS accelerated dimensionality reduction*, software repository (2026), https://github.com/sashakolpakov/dire-rapids.

[5] W. Dong, C. Moses, and K. Li, "Efficient k-nearest neighbor graph construction for generic similarity measures," in *Proceedings of the 20th International Conference on World Wide Web* (2011), pp. 577–586, https://doi.org/10.1145/1963405.1963487.

[6] RAPIDS cuML Developers, *UMAP API documentation, cuML 26.06*, https://docs.rapids.ai/api/cuml/stable/api/generated/cuml.manifold.umap/ (accessed July 2026).

[7] RAPIDS cuVS Developers, *All-neighbors and NN-descent API documentation, cuVS 26.06*, https://docs.rapids.ai/api/cuvs/stable/python_api/neighbors_all_neighbors/ (accessed July 2026).

[8] E. Facco, M. d'Errico, A. Rodriguez, and A. Laio, "Estimating the intrinsic dimension of datasets by a minimal neighborhood information," *Scientific Reports* 7 (2017), 12140.

[9] E. Levina and P. J. Bickel, "Maximum likelihood estimation of intrinsic dimension," in *Advances in Neural Information Processing Systems 17* (2004), pp. 777–784.

[10] J. Venna and S. Kaski, "Neighborhood preservation in nonlinear projection methods: an experimental study," in *Artificial Neural Networks—ICANN 2001*, pp. 485–491.

[11] U. Bauer, "Ripser: efficient computation of Vietoris–Rips persistence barcodes," *Journal of Applied and Computational Topology* 5 (2021), 391–423.

[12] S. Salvador and P. Chan, *Toward accurate dynamic time warping in linear time and space*, *Intelligent Data Analysis* 11 (2007), 561–580.

[13] Beijing Academy of Artificial Intelligence, *BGE small English v1.5*, https://huggingface.co/BAAI/bge-small-en-v1.5 (2023).

[14] Cohere Labs, *Wikipedia 2023-11 multilingual-v3 embeddings*, Hugging Face dataset card (accessed July 2026).

[15] A. Babenko and V. Lempitsky, "Efficient indexing of billion-scale datasets of deep descriptors," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2016), pp. 2055–2063.

[16] D. Baranchuk and A. Babenko, "Benchmarks for billion-scale similarity search," Yandex Research, 26 April 2021, https://research.yandex.com/blog/benchmarks-for-billion-scale-similarity-search.

[17] The Tabula Sapiens Consortium, "The Tabula Sapiens: a multiple-organ, single-cell transcriptomic atlas of humans," *Science* 376 (2022), eabl4896.

[18] Chan Zuckerberg Initiative, *CELLxGENE Census, data release 2025-11-08*, https://chanzuckerberg.github.io/cellxgene-census/ (accessed July 2026).

[19] I. Rivin, *dimension-reduction-benchmarks: autoencoder, DiRe, and UMAP experiments*, companion software and data repository (2026), https://github.com/igorrivin/dimension-reduction-benchmarks.